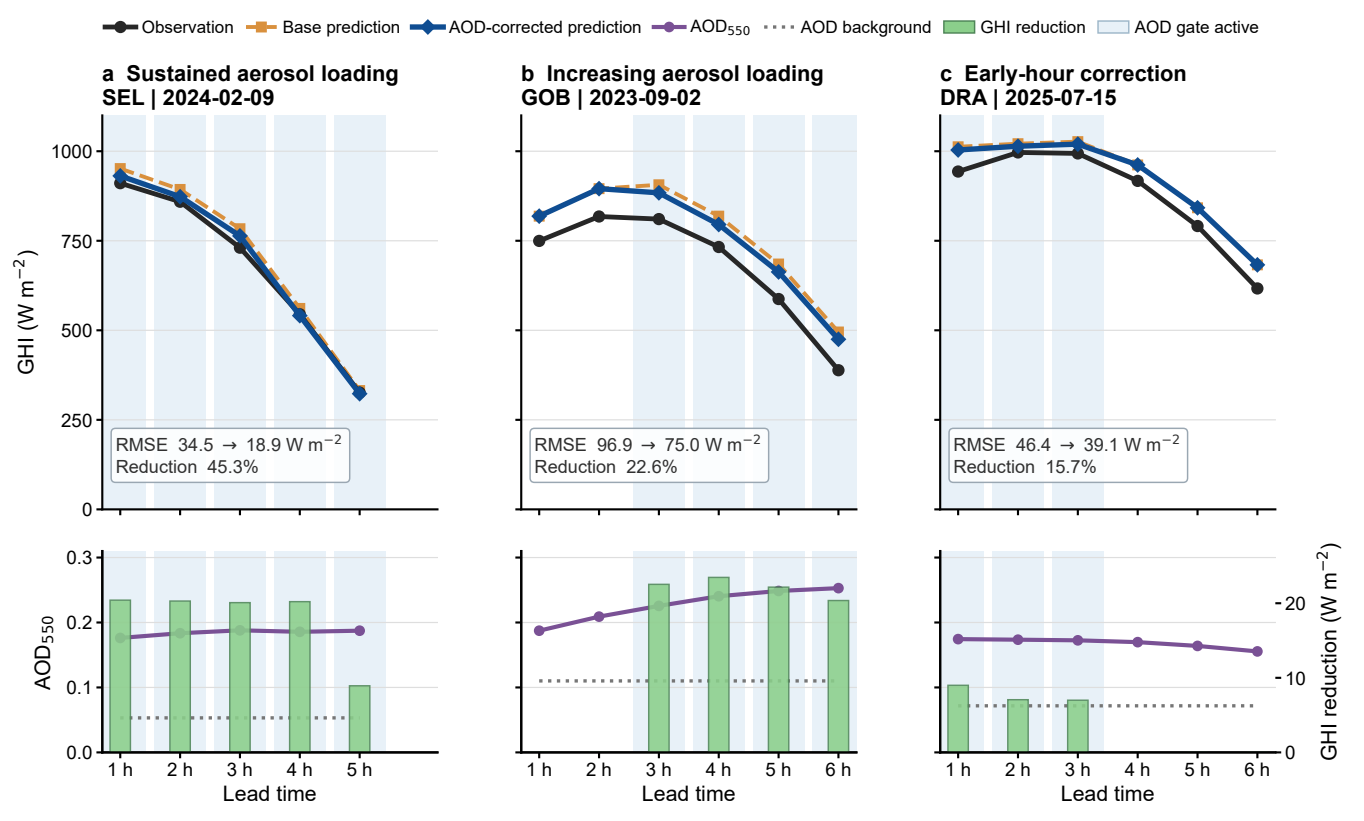

Observation
Base prediction
AOD-corrected prediction
AOD550
AOD background
GHI reduction
AOD gate active
a Sustained aerosol loading
SEL | 2024-02-09
b Increasing aerosol loading
GOB | 2023-09-02
c Early-hour correction
DRA | 2025-07-15
GHI (W m−2)
1000
750
500
250
0
RMSE 34.5 → 18.9 W m−2
Reduction 45.3%
RMSE 96.9 → 75.0 W m−2
Reduction 22.6%
RMSE 46.4 → 39.1 W m−2
Reduction 15.7%
AOD550
0.3
0.2
0.1
0.0
GHI reduction (W m−2)
20
10
0
1 h
2 h
3 h
4 h
5 h
6 h
Lead time